

GENERAL SPECIFICATION G7.19

Fire Pumps and Fire Water Storage

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G7.19 FIRE PUMPS AND FIRE WATER STORAGE

G7.19.1 Scope

- (a) This Section of the Specification contains requirements necessary to provide and install complete, fire water storage tanks, pumps, pressure maintenance (jockey) pumps, drivers, controllers, accessories, piping, and fittings.
- (b) The Fire Water Storage Tanks and associated fire pumping system will form the reliable water supply source for the proposed water-based firefighting system and fire protection system. These assets are specified to serve as water supply source for fire mains.
- (c) The Fire Water Storage Tanks will receive Domestic water supply feed from the PUB water mains with approved PUB bulk metering, piping, fittings and valves chamber facilities.
- (d) An indicative location of the Fire Water Storage Tanks is shown on the general arrangement proposed for the site on the Drawings.

G7.19.2 Standards

- (a) Work included in the Section shall conform to the following, except as specified otherwise within this Section:
 - (i) NFPA 13, Sprinkler Systems.
 - (ii) NFPA 20, Centrifugal Fire Pumps.
 - (iii) NFPA 22, Standard for Water Tanks for Private Fire Protection.
 - (iv) Singapore Civil Defence Force SCDF Fire Code 2013.
 - (v) Singapore Productivity and Standards Board (PSB):
 - 1) CP29, Code of Practice for Fire Hydrant Systems and Hose Reels.
 - 2) SSCP636, Code of Practice for Water Services.
 - 3) CP52, Code of Practice for Automatic Fire Sprinkler System.
 - 4) SS532, Code of Practice for The Storage of Flammable Liquids.
 - 5) SS575, Code of Practice for Fire Hydrant, Rising Mains and Hose reel Systems.
 - 6) Product Listing Scheme.
 - (vi) FMRC Data Sheet 3-7N/13-4N, Fire Pumps.
 - (vii) IRI Information Sheet IM.14.2.1, Centrifugal Fire Pumps.
 - (viii) BCA Code of Practice on Buildable Design / Buildability to the latest edition.
 - (ix) Ministry of Manpower- Code of Practice for Workplace Safety and Health (WSH).

G7.19.3 Definitions

- (a) Reference to a device or a part of the equipment applies to the number of devices or parts required to complete the installation.
- (b) Churn Pressure refers to pump discharge pressure at zero flow.

G7.19.4 Design Criteria

- (a) Performance:
 - (i) Fire Pump: adjust setting for start at a pressure within 0.7bar of churn pressure.
 - (ii) Relief Valve: valve relief pressure 0.3bar greater than churn pressure.
 - (iii) Stopping Accuracy: plus or minus maximum 5mm under any loading condition.
 - (iv) Jockey Pumps - Rated Capacity: size to exceed the flow from inspector's test valve without starting fire water pump.
- (b) Equipment shall be UL listed and FM approved for fire protection services.
- (c) Electrical driven sprinkler pump, jockey pump, and hose-reel pump shall be PSB approved and listed.
- (d) Contractor shall file related documents for permits for the portion of the fire sprinkler systems covered by the work, and arrange and pay for inspections and tests of the fire sprinkler system required by codes and standards and authorities having jurisdiction, other than the regular RI fees for testing.

G7.19.5 Regulatory Requirements

- (a) The fire system shall comply with the most stringent applicable provisions of the Codes, ordinances, and laws applicable within Singapore, including revisions and changes in effect on date of these Specifications.
- (b) The Contractor shall submit calculations which demonstrate compliance with requirements of specified key references and codes, such as NFPA 22 and AWWA D100 or equal to comply with AHJ (Authorities Having Jurisdiction).

G7.19.6 Submittals

- (a) The Contractor's drawings shall be endorsed by a Professional Engineer registered in Singapore who has proven experience in works of this nature. The plans shall be properly drawn to scale and prepared in a format suitable for submission to the Building and Construction Authority, Singapore and other relevant authorities.
- (b) Submit datasheets from coating manufacturer as part of submittal.
- (c) Provide the following submittals for the fire pump system:
 - (i) Detailed installation shop drawings of the complete pump system. Drawings shall show information required by NFPA 20 for working plans and include:
 - 1) Pumps and drivers, jockey pumps.
 - 2) Installation dimensions and cut lengths for piping.
 - 3) Valves, fittings, and accessories.

- 4) Sensing lines and controllers.
- (ii) Technical data sheet and pump curves for impeller efficiency, NPSH available, absorbed brake horsepower, rpm, exhaust stack and diesel tank size.
- (iii) Product data, including details on the pump, driver, controller, piping, and valving.
- (iv) Product certificates signed by manufacturer.
- (v) Bill of materials showing make, model, part number, and options.
- (vi) Controller wiring interconnection and schematic, motor horsepower, and power supply voltage.
- (vii) Full hydraulic calculation for most unfavourable and most favourable areas.
- (viii) Certified fire pump performance curve.
- (ix) Pump House Drawings: size, construction materials and drainage design. Operation and maintenance manuals, final drawings, and final calculations.
- (x) Operation and maintenance data.

G7.19.7 Products

- (a) General
 - (i) Manufacturers shall be listed in the latest published edition of Factory Mutual Research Corporation's Fire Protection Approval Guide.
- (b) Fire Pumps
 - (i) Pumps - Type:

Casing material	Cast Iron
Bearing housing material	Cast Iron
Casing bolts material	Steel
Bearing housing Gasket material	Rubber
Bearing housing bolts material	Steel
Bearing cap bolts material	Steel
Impeller material	Bronze
Impeller key material	Steel
Shaft material	Steel
Shaft sleeve material	Bronze
Shaft Seal Material	Graphited Acrylic Yarn
Motor	Suitable for Direct On Line Operation

- (ii) Drivers:
 - 1) Diesel Engine: water-cooled heat exchanger, and electric starter.
 - 2) Electric Motors: electric motor, totally enclosed. Comply with G5, OEM Electric Motors
- (iii) Accessories:
 - 1) Signal Contacts: Form C remote contacts, Fire Alarm and Smoke Detection Systems.
- (iv) Alarms:
 - 1) Diesel Driver:
 - Overheated engine.
 - Low and high pump room temperature.
 - Low fuel.
 - Suction tank low level.
 - Sequential timing device.
 - Protected against overcurrent of 150 percent.
 - Battery charger supply failure.
 - Failure of engine to start automatically.
 - Diesel start circuit disconnected.
 - 2) Electric Driver:
 - Low and high pump room temperature.
 - Sequential timing device.
 - Protected against overcurrent of 150 percent.
- (v) Fuel Tank for Diesel Fire Pump: Comply with CP 52.
- (vi) Fuel Piping: Comply with G4, Carbon Steel Piping Systems.
- (vii) Vibration isolation.
- (viii) Inlet and Exhaust System:
 - 1) Intake Air Filter: dry type.
 - 2) Spark Arrestor/Silencer: residential noise-rated, heavy industrial type with flexible exhaust connection.
 - 3) Sound Power Level: 45 dB(A) at 30m.
 - 4) Silencer: sized for the engine with ANSI flanges and rain cap.
 - 5) Stainless steel (SS304) exhaust pipe of size not less than the engine exhaust outlet shall be installed from the engine exhaust connection to high level outside the pump room. The exhaust pipe shall be provided with high temperature insulation.

- (ix) Batteries:
 - 1) Maintenance free lead-acid type comply with AS 4029.2 or AS.
 - 2) 4029.3 or be the nickel cadmium comply with AS 3731.
 - 3) Housed on brackets with 150mm above floor and fitted with non-conductive cover.
 - 4) Battery charger: Capable of fully recharging the battery from 50% capacity within 24 hours. Associated ammeter and voltmeter shall be incorporated into controller cabinet.
 - 5) System Piping - Use new materials that comply with the requirements of NFPA 13, including the following:
 - Ratings: between pump discharge flange and relief valve, 20bar and others at 12bar at 15 to 65 degrees C per ASTM A135, NFPA 13, and FMRC-2-8N, Table 3-1.1.1.
 - Piping:
 - 40mm and Smaller: ASTM A135, CW, Schedule 40, black, threaded.
 - 50mm and Larger: ASTM A135, CW, Schedule 40, black, grooved, or flanged end.
- (x) Plain-End Pipe Fittings and Couplings: prohibited, including setscrew couplings.
- (xi) Bushings and Grooved Reducing Couplings: prohibited.
- (xii) Uniflanges: prohibited.
- (c) Spare Parts:
 - (i) Provide one complete set of spare mechanical seals for each pump type.
 - (ii) Provide one complete set of spare bearings for each pump type.
- (d) Fire Water Tanks
 - (i) Fabricated of bolted pressed steel panels supported on reinforced concrete plinths or RC tank. Tank shall be constructed and comply with the requirement of CP 48 Code of Practise for Water Supplies.
 - (ii) Tank shall be compartmentalized into two (2) separate independent water-tight sections connected by valve and piping system.
 - (iii) Field-erected tanks shall include all necessary ladders, handrails, walkways, lugs, clips, bolts, nuts, gaskets, internal support brackets, and internal piping.
 - (iv) Tank bottoms shall be flat with false bottom, continuously sloped to drain.
 - (v) Nozzle projections for tanks to be at least 150mm. Projection for each pump suction or outlet nozzle and for each access opening or manway shall be 300mm.
 - (vi) Provide minimum two grounding lugs for M10 bolt connections.

- (vii) Provide and install 1 interior ribbon anode for passive cathodic protection.
- (viii) Each tank to be provided with level gauge / indicator assembly. Include stainless steel encased float with tape, guides, bottom guide anchor, two spring-loaded top guide anchors, two 90 degree sheave elbows, and interconnecting Schedule 40 galvanized carbon steel pipe.
- (ix) The fire water tank shall be provided with an approved type high and low level alarm system and interface with the solenoid/motor actuated valve. The valve will be closed during high water high level alarm. DC supply of this valve shall be taken from main alarm panel. This alarm system shall be installed at the fire pump control panel located within the fire pump room and main fire alarm panel. High and low level audible/ visual alarm indications, and audible alarm muting/ reset buttons shall be provided at the panel. As per specified key references, standards and documentation in accordance to fire code for compliance.

G7.19.8 Cleaning

- (a) Prior to shipping, perform final cleaning of internal and external surfaces after manufacturing and testing operations have been completed.

G7.19.9 Execution

- (a) Installation
 - (i) Install each equipment item in accordance with manufacturer's directions, referenced codes, and specifications.
 - (ii) Alignment Procedures shall comply with NFPA 20, appendix item A-3-5.
 - (iii) Install items so they may be easily removed for maintenance and repair.
 - (iv) Install items so that access for maintenance is safe and readily available.
 - (v) Fuel Tank Installation shall comply with NFPA 20, Figure A-8-4.6. Fuel Tank Low-Level Sensor alarm shall be set at minimum level allowed by NFPA 20.
- (b) Testing
 - (i) Acceptance Tests: Comply with NFPA 13 and 20, including NFPA 20, appendix items A-11-2.6.3 and A-11-2.6.3(f), Test Procedure, and requirements of the authority having jurisdiction.
 - (ii) Fire water tanks to be tested in place by sealing connections, filling with water, and allowing water to stand in tank for minimum of 24 hours. Any leaks including visible wet patches, or defects which may cause leakage, shall be rectified and re-tested until the tank is completely watertight. If a leak is detected while the tank is being filled with water, the Contractor may repair the defect responsible for the leak before continuing filling the tank to the top water level. Filling and emptying of the tank shall be carried out by the Contractor. A copy of the hydrostatic test results shall be submitted to the S.O.

END OF SECTION